1. Let $5f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}+3,\ x>0$. Then $18\int_1^2 f(x)dx$ is equal to

[2023 (06 Apr Shift 1)]

(1) $5\ \log_e 2+3$

(2) $10\ \log_e 2+6$

(3) $10\ \log_e 2-6$

(4) $5\log_e 2-3$

2. Let $A=\left\{x\in\mathbb{R}:[x+3]+[x+4]\leq 3\right\},\ B=\left\{x\in\mathbb{R}:3^x\left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3}<3^{-3x}\right\}$, where $[t]$ denotes greatest integer function. Then,

[2023 (06 Apr Shift 1)]

(1) $B\subset C,\ A\neq B$

(2) $A\cap B=\phi$

(3) $A\subset B, A\neq B$

(4) $A=B$

3. Let the sets A and B denote the domain and range respectively of the function $f(x)=\frac{1}{\sqrt{\lceil x\rceil-x}}$, where $\lceil x\rceil$ denotes the smallest integer greater than or equal to x. Then among the statements

$(S1)$: $A\cap B=(1,\ \infty)-\mathbb{N}$ and

$(S2)$: $A\cup B=(1,\ \infty)$

[2023 (06 Apr Shift 2)]

(1) Only $(S2)$ is true

(2) Only $(S1)$ is true

(3) Neither $(S1)$ nor $(S2)$ is true

(4) Both $(S1)$ and $(S2)$ are true

4. If domain of the function $\log_e\left(\frac{6x^2+5x+1}{2x-1}\right)+\cos^{-1}\left(\frac{2x^2-3x+4}{3x-5}\right)$ is $(\alpha,\beta)\cup(\gamma,\delta)$, then $18\left(\alpha^2+\beta^2+\gamma^2+\delta^2\right)$ is equal to

[2023 (08 Apr Shift 2)]

5. Let $R=\{a,b,c,d,e\}$ and $S=\{1,2,3,4\}$. Total number of onto functions $f:R\to S$ such that $f(a)\neq 1$, is equal to ________.

[2023 (08 Apr Shift 2)]

6. If the domain of the function $f(x)=\sec^{-1}\left(\frac{2x}{5x+3}\right)$ is $[\alpha,\beta)\cup(\gamma,\delta]$, then $|3\alpha+10(\beta+\gamma)+21\delta|$ is equal to __________

[2023 (10 Apr Shift 2)]

7. The domain of the function $f(x)=\frac{1}{\sqrt{[x]^2-3[x]-10}}$ is (where $[x]$ denotes the greatest integer less than or equal to x)

[2023 (11 Apr Shift 2)]

(1) $(-\infty,-3]\cup(5,\infty)$

(2) $(-\infty,-2)\cup[6,\infty)$

(3) $(-\infty,-2)\cup(5,\infty)$

(4) $(-\infty,-3]\cup[6,\infty)$

8. Let $A=\{1,2,3,4,5\}$ and $B=\{1,2,3,4,5,6\}$. Then the number of functions $f:A\to B$ satisfying $f(1)+f(2)=f(4)-1$ is equal to........

[2023 (11 Apr Shift 2)]

9. Let D be the domain of the function $f(x)=\sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$. If the range of the function $g:D\to\mathbb{R}$ defined by $g(x)=x-[x]$, ($[x]$ is the greatest integer function), is $(\alpha,\ \beta)$, then $\alpha^2+\frac{5}{\beta}$ is equal to

[2023 (12 Apr Shift 1)]

(1) 135

(2) 45

(3) 46

(4) 136

10. For $x\in\mathbb{R}$, two real valued functions $f(x)$ and $g(x)$ are such that, $g(x)=\sqrt{x}+1$ and $fog(x)=x+3-\sqrt{x}$. Then $f(0)$ is equal to

[2023 (13 Apr Shift 1)]

(1) 1

(2) 5

(3) 0

(4) -3

11. For the differentiable function $f: \mathbb{R}-\{0\} \rightarrow \mathbb{R}$, let $3 f(x)+2 f\left(\frac{1}{x}\right)=\frac{1}{x}-10$, then $\left|f(3)+f^{\prime}\left(\frac{1}{4}\right)\right|$ is equal to

[2023 (13 Apr Shift 1)]

(1) $\frac{33}{5}$
(2) 8
(3) $\frac{29}{5}$
(4) 13

12. The range of $f(x)=4 \sin ^{-1}\left(\frac{x^2}{x^2+1}\right)$ is

[2023 (13 Apr Shift 2)]

(1) $[0,2 \pi]$
(2) $[0, \pi]$
(3) $[0,2 \pi)$
(4) $[0, \pi)$

13. If the domain of the function $f(x)=\log _e\left(4 x^2+11 x+6\right)+\sin ^{-1}(4 x+3)+\cos ^{-1}\left(\frac{10 x+6}{3}\right)$ is $(\alpha, \beta]$, then $36|\alpha+\beta|$ is equal to

[2023 (15 Apr Shift 1)]

(1) 54
(2) 72
(3) 63
(4) 45

ANSWER KEYS

1. (3)	**2.** (4)	**3.** (3)	**4.** (20)	**5.** (180)	**6.** (24)	**7.** (2)	**8.** (360)
9. (1)	**10.** (2)	**11.** (4)	**12.** (3)	**13.** (4)			

1. **(3)**

Given:

$5f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}+3 \quad \ldots(1)$

Replace $x\rightarrow\frac{1}{x}$

$5f\left(\frac{1}{x}\right)+4f(x)=x+3 \quad \ldots(2)$

Eliminating $f\left(\frac{1}{x}\right)$ from (1) & (2), we get

$9f(x)=\frac{5}{x}+15-4x-12$

$\Rightarrow 9f(x)=\frac{5}{x}-4x+3$

$\Rightarrow 18f(x)=\frac{10}{x}-8x+6$

$\Rightarrow 18\int_1^2 f(x)dx=\int_1^2\left(\frac{10}{x}-8x+6\right)dx$

$\Rightarrow 18\int_1^2 f(x)dx=\left[10\log x-4x^2+6x\right]_1^2$

$\Rightarrow 18\int_1^2 f(x)dx=[(10\log 2-4)-(10\log 1+2)]$

$\Rightarrow 18\int_1^2 f(x)dx=10\log_e 2-6$

Hence this is the correct option.

2. **(4)**

Given,

$A=\left\{x\in\mathbb{R}:[x+3]+[x+4]\leq 3\right\}$ & $B=\left\{x\in\mathbb{R}:3^x\left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3}<3^{-3x}\right\}$,

Now solving A we get,

$[x+3]+[x+4]\leq 3$

$\Rightarrow [x]+3+[x]+4\leq 3$

$\Rightarrow 2[x]\leq -4$

$\Rightarrow [x]\leq -2$

$\Rightarrow x<-1\Rightarrow A=(-\infty,-1)$

Now solving B we get,

$3^x\left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3}<3^{-3x}$

Now using infinite G.P formula $S_\infty=\frac{a}{1-r}$ we get,

$\Rightarrow 3^x\left(\frac{3\cdot\frac{1}{10}}{1-\frac{1}{10}}\right)^{x-3}<3^{-3x}$

$\Rightarrow 3^x\left(\frac{1}{3}\right)^{x-3}<3^{-3x}$

$\Rightarrow 3^{x-x+3+3x}<1$

$\Rightarrow 3^{3(x+1)}<1$

$\Rightarrow 3(x+1)<0$

$\Rightarrow x<-1\Rightarrow B=(-\infty,-1)$

$\Rightarrow A=B$

3. (3)

Given,

$f(x) = \frac{1}{\sqrt{[x]-x}}$

We know that, $x = [x] + \{x\}$ where $\{x\}$ is fractional part of x whose value is $0 < \{x\} < 1$,

So, $[x] - x = -\{x\}$

Now putting the value in given function we get,

$f(x) = \frac{1}{\sqrt{[x]-x}}$

$\Rightarrow f(x) = \frac{1}{\sqrt{-\{x\}}}$

Now we know that $\frac{1}{\sqrt{g(x)}}$ is defined when $g(x) > 0$, but $-\{x\} \in (-1, 0)$

Hence, $\sqrt{-\{x\}}$ is not defined

Hence, Domain and Range will be $= \phi$

4. (20)

We need to find the domain of the function $\log_e\left(\frac{6x^2+5x+1}{2x-1}\right) + \cos^{-1}\left(\frac{2x^2-3x+4}{3x-5}\right)$.

$\Rightarrow \frac{6x^2+5x+1}{2x-1} > 0 \ldots\ldots (1)$ and

$\Rightarrow \ -1 \leq \frac{2x^2-3x+4}{3x-5} \leq 1 \ldots\ldots (2)$

Now From (1),

$\Rightarrow \frac{6x^2+5x+1}{2x-1} > 0$

$\Rightarrow \frac{(3x+1)(2x+1)}{(2x-1)} > 0$

Now we get the common region here as

$\Rightarrow x \in \left(-\frac{1}{2}, -\frac{1}{3}\right) \cup \left(\frac{1}{2}, \infty\right) \ldots\ldots\ldots (\mathrm{a})$

From (2) we get that

$\Rightarrow -1 \leq \frac{2x^2-3x+4}{3x-5} \leq 1$

$\Rightarrow \frac{2x^2-3x+4}{3x-5} \geq -1 \ldots\ldots (3)$

and $\frac{2x^2-3x+4}{3x-5} \leq 1 \ldots\ldots (4)$

From (3) we get

$\Rightarrow \frac{2x^2-3x+4}{3x-5} + 1 \geq 0$

$\Rightarrow x \in \left[\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right] \cup \left(\frac{5}{3}, \infty\right) \ldots\ldots (\mathrm{b})$

From (4) we get

$\Rightarrow \frac{2x^2-3x+4}{3x-5} - 1 \leq 0$

$\Rightarrow \frac{2x^2-6x+9}{3x-5} \leq 0$

As $D = (-6)^2 - 4(2)(9) < 0$ and $2 > 0$, we observe that $2x^2 - 6x + 9 > 0\ \forall x \in \mathbb{R}$

$\Rightarrow \frac{1}{3x-5} \leq 0\ \ (\because 2x^2 - 6x + 9 > 0 \forall x \in R)$

$\Rightarrow \ \ x \in \left(-\infty, \frac{5}{3}\right) \ \ \ldots\ldots (\mathrm{c})$

$\Rightarrow$ Intersection of (a), (b) and (c) gives us

$x \in \left(-\frac{1}{2}, -\frac{1}{3}\right) \cup \left(\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$.

On comparing this with $(\alpha, \beta) \cup (\gamma, \delta)$,

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 + \delta^2 = \frac{10}{9}$

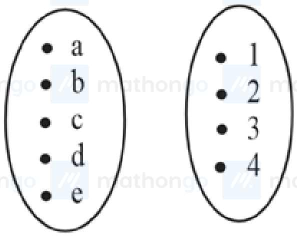
$\Rightarrow 18\left(\alpha^2 + \beta^2 + \gamma^2 + \delta^2\right) = 20.$

Hence this is the required answer.

5. (180)

Given,

$R=\{a, b, c, d, e\}$ and $S=\{1,2,3,4\}$



Now taking, $f(a)=1$ we get, one of $f(b),\ f(c),\ f(d),\ f(e)=1$ then total such cases $=4 \cdot 3!=24$

Now if, only $f(a)=1$, then we have distribute $\{2,3,4\}$ amongst $\{b, c, d, e\}$,

So, total cases $=3^4-\left({ }^3 C_1 \cdot 2^4\right)+\left({ }^3 C_2 \cdot 1\right)$

$=36$

So, number of onto functions when $f(a)=1$ is $24+36=60$

Now finding, total number of onto functions,

$=4^5-\left({ }^4 C_1 \cdot 3^5\right)+\left({ }^4 C_2 \cdot 2^5\right)-\left({ }^4 C_3 \cdot 1\right)$

$=1024-973+192-4$

$=240$

Number of required functions when $f(a) \neq 1$ will be,

$=240-60=180$

6. (24)

We know that,

Domain of $\sec^{-1} x$ is $(-\infty,-1] \cup[1, \infty)$.

Now, $f(x)=\sec^{-1}\left(\frac{2 x}{5 x+3}\right)$

So,

$\frac{2 x}{5 x+3} \leq-1$ or $\frac{2 x}{5 x+3} \geq 1$

$\Rightarrow \frac{2 x}{5 x+3}+1 \leq 0$ or $\frac{2 x}{5 x+3}-1 \geq 0$

$\Rightarrow \frac{7 x+3}{5 x+3} \leq 0$ or $\frac{-3 x-3}{5 x+3} \geq 0$

$\Rightarrow \frac{7 x+3}{5 x+3} \leq 0$ or $\frac{x+1}{5 x+3} \leq 0$

$\Rightarrow x \in\left[-1,-\frac{3}{5}\right) \cup\left(-\frac{3}{5},-\frac{3}{7}\right]$

So,

$\alpha=-1,\ \beta=-\frac{3}{5},\ \gamma=-\frac{3}{5},\ \delta=-\frac{3}{7}$

Now,

$|3 \alpha+10(\beta+\gamma)+21 \delta|$

$=|-3-6-6-9|$

$=24$

Hence this is the required answer.

7. (2)

Given:

$f(x)=\frac{1}{\sqrt{[x]^2-3[x]-10}}$

For $f(x)$ to be defined,

$[x]^2-3[x]-10>0$

$\Rightarrow([x]-5)([x]+2)>0$

$\Rightarrow[x]<-2$ & $[x]>5$

$\Rightarrow x<-2$ & $x \geq 6$

So, domain is $(-\infty,-2) \cup[6, \infty)$.

8. (360)

Given,

$f(1)+f(2)+1=f(4)$

$\Rightarrow f(1)+f(2) \leq 5$ {as $f(4) \leq 6$}

Now taking cases we get,

Case (1), $f(1)=1 \Rightarrow f(2)=\{1,2,3,4\} \rightarrow 4$ mappings

Case (2), $f(1)=2 \Rightarrow f(2)=\{1,2,3\} \rightarrow 3$ mappings

Case (3), $f(1)=3 \Rightarrow f(2)=\{1,2\} \rightarrow 2$ mappings

Case (4), $f(1)=4 \Rightarrow f(2)=\{1\} \rightarrow 1$ mappings

And $f(5)$ & $f(6) \rightarrow 6$ mapping each

So, total number of function will be $=(4+3+2+1) \times 6 \times 6=360$

9. (1)

We have been given

$f(x)=\sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$

We know that the domain of $\sin^{-1} x$ is $-1 \leq x \leq 1$.

$\Rightarrow \underset{-(1)}{\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right) \in [-1\ \ 1]},\ \underset{-(2)}{3x>0},\ \underset{-(3)}{3x \neq 1},\ \underset{-(4)}{\frac{6+2\log_3 x}{-5x}>0}$

$\Rightarrow$ From (2), $x>0$, from (3), $x \neq \frac{1}{3}$

From (4), $6+2\log_3 x<0 \quad (\because\ x>0)$

$\Rightarrow \log_3 x<-3$

$\Rightarrow x<3^{-3}$

$\Rightarrow x<\frac{1}{27}$

From (2), (3), (4), $0<x<\frac{1}{27} \quad \ldots(5)$

From (1), $-1 \leq \log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right) \leq 1$

$\because\ 3x \in\left(0, \frac{1}{9}\right)$

$\Rightarrow \frac{1}{3x} \geq \frac{6+2\log_3 x}{-5x} \geq 3x$

$\Rightarrow -\frac{5}{3} \leq 6+2\log_3 x \leq -15x^2$

$\Rightarrow -\frac{23}{6} \leq \log_3 x \leq \frac{-15x^2-6}{2}$

$\Rightarrow 3^{-\frac{23}{6}} \leq x \leq 3^{\frac{-15x^2-6}{2}}$

$\Rightarrow \frac{1}{3^{\frac{23}{6}}} \leq x \leq \frac{1}{3^{\frac{15x^2+6}{2}}}$

$\Rightarrow 3^{\frac{-23}{6}} \leq x<\frac{1}{3^{\frac{15x^2+6}{2}}} \quad \ldots\ldots(6)$, here $3^{\frac{-23}{6}}$ is very small positive quantity,

From (5) and (6), $3^{\frac{-23}{6}}<x<\frac{1}{27}$, $[x]=0$

$\Rightarrow g(x)=x$

$\Rightarrow$ Range $g(x)$ is domain of $f(x)$

$\Rightarrow g(x) \in\left(3^{\frac{-23}{6}}, \frac{1}{27}\right) \equiv(\alpha, \beta)$.

$\Rightarrow \alpha^2+\frac{5}{\beta}=\left(3^{\frac{-23}{6}}\right)^2+\frac{5}{\frac{1}{27}}=\left(3^{\frac{-23}{6}}\right)^2+135 \approx 135$

Note: This question was bonus in Jee Main April 2023 session.

10. (2)

We have been given that

$f(g(x))=x+3-\sqrt{x}$

$\Rightarrow f(\sqrt{x}+1)=x+3-\sqrt{x}$

Put $\sqrt{x}+1=t$

$\sqrt{x}=t-1$

$x=(t-1)^2$

$\Rightarrow f(t)=(t-1)^2+3-(t-1)$

$\Rightarrow f(0)=1+3+1=5$

Hence this is the correct option.

11. (4)

Given:

$3f(x)+2f\left(\frac{1}{x}\right)=\frac{1}{x}-10 \quad ....(1)$

Replace $x \rightarrow \frac{1}{x}$

$3f\left(\frac{1}{x}\right)+2f(x)=x-10 \quad ....(2)$

Eliminating $f\left(\frac{1}{x}\right)$ from (1) & (2), we get

$\Rightarrow 5f(x)=\frac{3}{x}-2x-10$

Let us replace x with 3

$\Rightarrow 5f(3)=\frac{3}{3}-2(3)-10$

$\Rightarrow f(3)=-3$

$\Rightarrow 5f'(x)=\frac{-3}{x^2}-2$

$\Rightarrow 5f'\left(\frac{1}{4}\right)=-48-2$

$\Rightarrow f'\left(\frac{1}{4}\right)=-10$

$\Rightarrow \left|f\left(3\right)+f'\left(\frac{1}{4}\right)\right|=13$

Hence this is the correct option.

12. (3)

Given,

$f(x)=4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$

Now taking, $\frac{x^2}{1+x^2}=1-\frac{1}{1+x^2}<1$

$\Rightarrow 0 \leq \frac{x^2}{1+x^2}<1$

Now taking $\sin^{-1}$ we get,

$\Rightarrow 0 \leq \sin^{-1}\left(\frac{x^2}{1+x^2}\right)<\frac{\pi}{2}$

$\Rightarrow 0 \leq 4\sin^{-1}\left(\frac{x^2}{1+x^2}\right)<2\pi$

Hence, the range of $4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is $[0, 2\pi)$

13. (4)

Given that $f(x)= \log\left(4x^2+11x+6\right)+\sin^{-1}(4x+3)+\cos^{-1}\left(\frac{10x+6}{3}\right)$

We know that the domain of $A(x)+B(x)+C(x)$ is $D(A)\cap D(B)\cap D(C)$.

Let us find the domain of $\log\left(4x^2+11x+6\right)$.

We know that domain of $\log(f(x))$ is $f(x)>0$.

$\Rightarrow 4x^2+11x+6>0$

$\Rightarrow 4x^2+8x+3x+6>0$

$\Rightarrow (4x+3)(x+2)>0$

$\Rightarrow x\in(-\infty,-2)\cup\left(\frac{-3}{4},\infty\right)$

Domain of $\log\left(4x^2+11x+9\right)$ is $x\in(-\infty,-2)\cup\left(\frac{-3}{4},\infty\right)$.

Let us find the domain of $\sin^{-1}(4x+3)$

$\Rightarrow -1\le 4x+3\le 1$

$\Rightarrow -1\le x\le \frac{-1}{2}$

$\Rightarrow x\in\left[-1,\frac{-1}{2}\right]$

Let us find the domain of $\cos^{-1}\left(\frac{10x+6}{3}\right)$.

$\Rightarrow -1\le \frac{10x+6}{3}\le 1$

$\Rightarrow -3\le 10x+6\le 3$

$\Rightarrow \frac{-9}{10}\le x\le \frac{-3}{10}$.

Now the common domain is $\frac{-3}{4}\le x\le \frac{-1}{2}$.

$\Rightarrow x\in\left[\frac{-3}{4},\frac{-1}{2}\right]$

But given that the domain of $f(x)$ is $x\in[\alpha,\beta]$.

$\Rightarrow \alpha=\frac{-3}{4},\ \beta=\frac{-1}{2}$.

$\Rightarrow|36(\alpha+\beta)|=\left|36\left(\frac{-3}{4}+\left(\frac{-1}{2}\right)\right)\right|$

$\Rightarrow|36(\alpha+\beta)|=\left|36\times\frac{-5}{4}\right|$

$\Rightarrow|36(\alpha+\beta)|=45$

Therefore, the required answer is 45.